

SynDiff Synthetic Benchmark — Live Progress

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300/942 jobs complete (31.8%)

$S = I(X_1; Y | X_2) - I(X_1; Y)$ in nats; $S > 0$ synergy-dominant, $S < 0$ redundancy-dominant. SynDiff (ours) vs prior estimators.

Sweep progress

Phase	Method	Done	Expected
P1	batch_pid	0	96
P1	broja	0	6
P1	infonce	75	96
P1	minde_comp	72	96
P1	mine	75	96
P1	plugin	2	6
P1	syndiff	76	96
P2	batch_pid	0	90
P2	infonce	0	90
P2	minde_comp	0	90
P2	mine	0	90
P2	syndiff	0	90

Accuracy vs ground truth (completed cells, lower error better)

Method	#cells	mean $ \text{est} - \text{truth} $	sign acc
minde_comp	25	0.0180	0.96
syndiff	26	0.0307	0.96
plugin	2	0.1343	1.00
mine	25	0.7438	0.56
infonce	25	5.8812	0.24

SynDiff vs prior — head to head (nats)

Cell	truth	SynDiff	batch_pid	minde_comp	mine	infonce	plugin
p1_dial_rho0_sy0p1_d8_n20k	+1.963	+1.887	—	+1.922	+0.631	-5.604	—
p1_dial_rho0_sy1_d8_n20k	+0.144	+0.136	—	+0.145	-0.555	-5.315	—
p1_dial_rho0p3_sy0p1_d8_n20k	+1.740	+1.677	—	+1.678	+0.627	-5.460	—
p1_dial_rho0p3_sy1_d8_n20k	+0.007	+0.006	—	+0.010	-0.603	-5.429	—
p1_dial_rho0p5_sy0p1_d8_n20k	+1.477	+1.407	—	+1.434	+0.525	-5.436	—
p1_dial_rho0p5_sy1_d8_n20k	-0.134	-0.134	—	-0.128	-0.829	-5.254	—
p1_dial_rho0p7_sy0p1_d8_n20k	+1.035	+0.971	—	+0.991	+0.309	-5.395	—
p1_dial_rho0p7_sy1_d8_n20k	-0.329	-0.334	—	-0.321	-0.651	-5.190	—
p1_dial_rho0p9_sy0p1_d8_n20k	+0.024	-0.032	—	-0.025	-0.329	-5.321	—
p1_dial_rho0p9_sy1_d8_n20k	-0.610	-0.601	—	-0.612	-0.807	-5.145	—

Cell	truth	SynDiff	batch_pid	minde_comp	mine	infonce	plugin
p1_dial_rhon0p3_sy0p1_d8_n20k	+2.047	+1.976	–	+2.017	+0.724	-5.385	–
p1_dial_rhon0p3_sy1_d8_n20k	+0.209	+0.199	–	+0.210	-0.562	-5.363	–
p1_dial_rhon0p5_sy0p1_d8_n20k	+2.023	+1.953	–	+1.980	+0.795	-5.507	–
p1_dial_rhon0p5_sy1_d8_n20k	+0.213	+0.197	–	+0.214	-0.504	-5.342	–
p1_dial_rhon0p7_sy0p1_d8_n20k	+1.896	+1.826	–	+1.853	+0.689	-5.488	–
p1_dial_rhon0p7_sy1_d8_n20k	+0.177	+0.158	–	+0.182	-0.648	-5.374	–
p1_dial_rhon0p9_sy0p1_d8_n20k	+1.473	+1.412	–	+1.445	+0.466	-5.470	–
p1_dial_rhon0p9_sy1_d8_n20k	+0.083	+0.054	–	+0.089	-0.580	-5.200	–
p1_ns_rho0p7_sy1_d8_n100k	-0.329	-0.332	–	–	–	–	–
p1_ns_rho0p7_sy1_d8_n1k	-0.329	-0.331	–	-0.325	-0.630	-5.217	–
p1_ns_rho0p7_sy1_d8_n20k	-0.329	-0.333	–	-0.324	-0.652	-5.261	–
p1_ns_rho0p7_sy1_d8_n5k	-0.329	-0.331	–	-0.324	-0.651	-5.174	–
p1_ns_rhon0p7_sy1_d8_n100k	+0.177	+0.157	–	+0.180	-0.648	-5.374	–
p1_ns_rhon0p7_sy1_d8_n1k	+0.177	+0.160	–	+0.183	-0.497	-5.338	–
p1_ns_rhon0p7_sy1_d8_n20k	+0.177	+0.154	–	+0.180	-0.602	-5.375	–
p1_ns_rhon0p7_sy1_d8_n5k	+0.177	+0.154	–	+0.185	-0.453	-5.449	–